Xiaoxiao Tu

xiaoxiaotu@link.cuhk.edu.cn | github.com/tuxiaoxia0

Research Interests

Reinforcement learning, supervised fine-tuning, LLM agents

Education

The Chinese University of Hong Kong, Shenzhen (GPA: **3.922/4.00**, Rank **5/231**) Sept. 2024 – Jun. 2028

B.Eng. in Computer Science and Engineering

- Selected coursework: Machine Learning (A), Advanced Machine Learning (A), Introduction to Computer Systems (A), Optimization (A), Linear Algebra (A).

Research Experiences

Undergraduate Research in Agent Evaluation May 2026 – Present

***National University of Singapore**; Supervised by Prof. Doudou Zhou and Prof. Linjun Zhang*

- Designing semiparametrically efficient estimators that calibrate LLM judges against a limited human-annotated sample.

Undergraduate Research in LLM Post-Training Mar. 2025 – May 2026

***CUHK-Shenzhen**; Supervised by Prof. Ruoyu Sun and Prof. Zhi-Quan (Tom) Luo*

- Improving parameter-efficient fine-tuning and reinforcement learning for reasoning LLMs.
- Building LLM agents that plan and verify numerical search, with one work at the **ICML 2026 AI4Math Workshop**.

Selected Research Projects

Calibrating LLM Judges for Statistically Efficient Agent Evaluation May 2026 – Present

Supervised by Prof. Doudou Zhou and Prof. Linjun Zhang

- Observed that agent evaluation increasingly relies on LLM-as-a-judge because human annotation is too costly to collect at scale, yet judges are unreliable on the long-horizon trajectories these tasks produce.
- Proposed a **debiased M-estimator** that treats multi-dimensional information extracted from each trajectory as an auxiliary signal and uses **cross-fitting** to correct judge verdicts toward the human-label scale.
- Derived the **efficient influence function** for a general estimating function, which yields asymptotic normality, the **semiparametric efficiency bound** and strictly lower variance than the naive estimator. Writing accuracy, precision, recall and F1 as affine estimating functions extends the same guarantee beyond accuracy.
- Cut absolute estimation error on AgentRewardBench by **2.66, 2.58 and 1.80 percentage points** on recall, accuracy and F1 over the naive sample mean at the same annotation budget.

Spectral Warm-Start for LoRA: Line-Searching Singular Values before Low-Rank Adaptation Mar. 2026 – Present

Supervised by Prof. Zhi-Quan (Tom) Luo and Prof. Ruoyu Sun

- Observed that the diagonal direction can be dropped from LoRA's fitted update without hurting quality.
- Proposed to train that diagonal on the **SVD of each pretrained weight** before adaptation starts, so the spectrum absorbs the large-magnitude information.
- Cut eval loss on Qwen3-0.6B **from 0.4290 to 0.4137** on MathInstruct, against 0.4108 for full fine-tuning.

Precision-Aware Automated Discovery of Numerical Function Approximations via Landscape-Adaptive Search and LLM Agents (ICML 2026 AI4Math Workshop) Aug. 2025 – Mar. 2026

Supervised by Prof. Zhi-Quan (Tom) Luo and Prof. Ruoyu Sun

- Co-built the outer **LLM agent** that turns a natural-language accuracy request into a verified numerical kernel, partitioning the input domain and dispatching searches through a typed tool interface.

Hinted-GRPO: Answer-Guided Reinforcement Learning for Diffusion-Based LLMs Nov. 2025 – Jan. 2026

Supervised by Prof. Zhi-Quan (Tom) Luo and Prof. Ruoyu Sun

- Proposed a fully on-policy **GRPO** variant for diffusion LLMs that exploits any-order denoising to inject the ground-truth answer as temporary guidance. **Answer-conditioned inpainting** elicits high-quality reasoning traces and **dynamic answer re-masking** then strips the hint.

Selected Awards and Honors

Full Tuition Scholarship offered by CUHK-Shenzhen, ∼$20,000 per year 2024 – 2028
Dean's List offered by CUHK-Shenzhen School of Data Science 2025
1st Place Winner, The 10th China Collegiate Programming Contest (CCPC), Women's National Finals 2025
Gold Medalist, The 49th International Collegiate Programming Contest (ICPC), Shanghai Regional 2024
Silver Medalist, The National Olympiad in Informatics (NOI) 2023
Silver Medalist, The Asia-Pacific Informatics Olympiad (APIO) 2023
Gold Medalist, The National Olympiad in Informatics Winter Camp (NOIWC) 2023

Teaching and Services

Undergraduate Student Teaching Fellow (USTF), CUHK-Shenzhen School of Data Science Sept. 2024 – Dec. 2024

CSC1003: Introduction to Computer Science and Java Programming

- Led tutorials and supported students completing their assignments.
- Delivered volunteer review lectures for CSC1003 before the final examination.